

MINDCRAFT

Research Handoff Summary

A product-focused digest of the Research Constitution: competitive landscape, pricing, metrics, and the SAFE-* product rules that survived Red Team.

Source: 160 chapters, 2026-07-25 to 2026-08-17, ~4,756 words in this digest.

MindCraft Research Handoff Summary

What this is: A standalone, product-focused digest of the "Research Constitution" — a research lab pipeline (Cursor background agent, multiple simulated researcher/red-team roles) that ran continuously from **2026-07-25 to 2026-08-17**, then stopped (the Cursor agent session was paused/lost and did not resume). It produced 160 chapters answering one question — *"How do humans become confident mathematical thinkers?"* — under one standing constraint: **every chapter must end in a product implication** (copy / feature / metric / kill), not academic filler.

Where this came from: agent_work/research/MINDCRAFT_RESEARCH_CONSTITUTION_v1.md (311KB core synthesis doc) + agent_work/research/chapters/*.md (140+ deep-dive chapters) + agent_work/research/RESEARCH_LOG.md (tick-by-tick log) + agent_work/research/CHAPTER_MANIFEST.txt (full chapter list).

Go deeper: If a topic below matters for what you're building, the cited chapter file (chapters/NN_slug.md) has full citations, confidence tables, and experiment designs the lab spawned. The Constitution's own table of contents (grep ^# in MINDCRAFT_RESEARCH_CONSTITUTION_v1.md) is the fastest way to find a specific Part.

Note: docs/canon/PEDAGOGY.md is the intended agent-facing digest of this research but is stale (last updated 2026-08-11, cites only through ~Part XXI, 47/159 SAFE-* rules) — this document is a separate, more current handoff, not a replacement for a future PEDAGOGY.md refresh.

How to read labels in this doc: The lab tags every claim FACT / HYPOTHESIS / FOUNDER BELIEF / SPECULATION. This summary preserves those labels wherever a specific claim is cited. Where a label is dropped it's because the sentence is describing what the *lab said*, not asserting the claim as true.

1. Core Thesis & North Star (already established — reused, not re-derived)

Core hypothesis (HYPOTHESIS): Confident math thinkers require three compounding things: **affective permission**, **competence evidence** (including self-accounted strategy, not just blocked accuracy), and **identity re-storying** (Eccles expectancy $\times$ value $\times$ cost). Fluent AI explanations alone can accelerate the *wrong* kind of confidence.

Working North Star (HYPOTHESIS): *"Challenge-seeking under safety with transfer"*

— a student picks a harder problem for mastery reasons, returns after failure

(retry_120s), and transfers on delayed/mixed sets (transfer_pass,

solo_transfer_pass). Explicitly **not** streaks, DAU, confidence-alone, or blocked accuracy.

Scarcity thesis (surviving form, Part XVII): Explanations are cheapening toward free. That does not make tutoring free — it relocates the product's value to **diagnosis + affect + identity + accountability**, with AI as infrastructure, not the product.

Scale: 159 named SAFE- * doctrines exist, each one a specific plausible-sounding bad product/metric idea (usually a vanity "___ Score™" or a shallow copy of a real practice) that got red-teamed and killed, replaced by a named discipline. Section 5 below samples ~35 of the broadest, most reusable ones.

2. Competitive Landscape

This is the section with the most concrete, citable findings — real papers, real effect sizes, real verdicts on named competitors. Primary sources: **Part XX** (Competitive Landscape, mechanism lens) and **Part XXVII** (`chapters/27\_markets\_parents\_competition.md`) for market positioning, and **Part XXXV** (chapters/35_competitive_session_audits.md) for session-level teardown — the chapter whose explicit "commercial job" was *"replace vague 'we're different' with a session-level wedge parents and students can feel in under five minutes."*

2.1 The mechanism table (Part XX)

Competitor	What they actually sell	Identity claim?	MindCraft's stated risk if it copies them
Khan Academy	Free mastery content + reputation	Weak–medium	Explanation commodity; "mastery" without transfer
Duolingo	Habit / streak motivation	Weak (sometimes language identity)	Extrinsic ceiling; consistency without depth
Brilliant	Aesthetic problem joy + prestige	Medium	Narrow segment; not MindCraft's core "Maya" (anxious/threatened) persona
Coursera/edX	Credentials	Medium (career identity)	Completion collapse

Competitor	What they actually sell	Identity claim?	MindCraft's stated risk if it copies them
Chegg / homework help	Answers under deadline	Anti-identity (outsourcing)	Integrity risk; AI-shocked business model
Private tutors	Human accountability + customization	High when good	Supply-constrained, not a moat MindCraft can out-scale on price
ChatGPT / generic AI tutors	Instant explanation	Low unless wrapped	Trust / hallucination; "Bastani Base" harm pattern (below)
MindCraft (target)	FEI conversion + tutor witness + gap diagnosis + solo transfer	Intended high	Must <i>prove</i> this, not assert it

Strategic implication (v1.20, condensed): Do not try to out-Khan Khan on content breadth, out-ChatGPT on unlimited full-worked solutions, out-Duo Duo on streaks/leagues/XP, or out-Brilliant Brilliant on puzzle delight. Compete on the **session + ops + honesty stack**: inspectable diagnosis, honest cold-start (no fake day-one mastery), format-gap conversion evidence, witnessable dated solo-competence artifacts, coverage matrices that name their own gaps, and faded guidance that actually earns solo carry — then *say exactly that* in marketing, nothing more.

2.2 Khan Academy — real dosage data, not just brand

FACT (large observational study, real classrooms): Eames, Brunskill, Yamkovenko, Weatherholtz & Oreopoulos (2026, *PNAS*) — panel of **>200,000 students** in districts using Khan's MAP Accelerator. Using within-teacher/within-school variation, they estimate $\sim +0.031$ SD math gain at observed average usage (~ 6.6 h/year $\approx$ 11 min/week), projecting to $\sim +0.085$ SD at the recommended ~ 30 min/week. Higher achievers benefit more.

FACT (null result): Kelly & Rutherford (2017, *IRRODL*) — a controlled study of Khan as supplemental instruction found **no significant difference** vs. control.

FACT (Khanmigo, mixed/early): A Puerto Rico pilot and an undergrad mixed-methods study both found qualitative receptivity but **no significant learning advantage** for Khanmigo vs. plain search/paper in short exposures.

Verdict (HYPOTHESIS, confidence High/Medium/Low as marked): Khan's session optimizes **coverage + dosage**, not affect or identity — dosage→modest-SD gains is High confidence; that an anxious student feels *seen* in a default Khan session is Medium; that Khanmigo currently delivers MindCraft-grade affective+evidence+identity (FEI) support without a human wrap is Low.

Commercial wedge: Don't out-library Khan. Parent copy tested by the lab: *"*Minutes on skills matter (Khan proved dosage). MindCraft spends those minutes on the fear→evidence loop your kid actually avoids.*"*

2.3 Duolingo — a world-class habit machine, the wrong North Star

FACT: Settles & Meeder (2016, ACL) — Duolingo's half-life regression (HLR) spacing model beat Leitner/Pimsleur baselines and produced ~+12% daily-activity retention in production A/B when it replaced the older scheduler.

FACT: A natural-experiment line on frequency/duration/intensity found total **minutes** correlated with written but *not* oral gains — frequency and curriculum-oriented intensity were more reliable proficiency correlates than raw time. Habit duration alone is not a sufficient statistic for learning.

Verdict: Duo wins the "open the app" problem (High confidence) — but habit ≠ FEI outcomes (High confidence). MindCraft should borrow on-ramp friction-reduction **without** colonizing meaning with streak/fire-icon mechanics (standing ban, Kill #11 below).

Commercial wedge tested: *"*We want you back — because something hard became yours, not because a streak dies at midnight.*"*

2.4 Brilliant — real craft, wrong core segment

FACT (small QED, Colombia, n=60): de la Puente & Perez (2023) found significant post-test/grade advantages for a Brilliant.org arm on linear algebra vs. non-Brilliant resources — but small sample, single topic, single context; not an anxiety- or identity-stratified design.

Verdict (HYPOTHESIS): Brilliant's session is the closest competitor to MindCraft on *active struggle* and anti-answer-dumping (its AI tutor "Koji" is explicitly constrained to fade before test-like moments). But it selects for students who already approach math as play — **not** MindCraft's primary threat-profile persona. Confidence is Low–Medium that the Colombia result generalizes to U.S. ACT-prep identity outcomes.

Commercial wedge tested: *"*Brilliant for curiosity already online. MindCraft for the kid who goes quiet when the room gets evaluative.*"*

2.5 ChatGPT / generic AI tutors — the single most load-bearing finding

FACT (field RCT, ~1,000 HS math students): Bastani, Bastani, Sungu, Ge, Kabakc■ & Mariman (2025, *PNAS*) — **GPT Base** (ChatGPT-like, unguarded) raised practice grades ~+48%; a pedagogy-guarded "**GPT Tutor**" (hint-not-answer + teacher keys embedded in the prompt) raised them ~+127% — during access. **When AI access was removed, GPT Base students scored ~17% worse on the exam than students who never had AI at all. * *GPT Tutor* neutralized* this harm (exam $\approx$ control) but the authors are explicit: **no positive learning effect vs. no-AI**. Mechanism: unguarded chat functions as a **crutch**, not only a hallucination risk.

This is the chapter's single sharpest correction of a natural founder assumption

(explicitly Red Teamed and partially killed, Part XXXIII.7a):

- **KILLED:** "Bastani's GPT Tutor is proof that MindCraft's ontology/pedagogy-wrap is the product." This is a **category error** — Bastani's "Tutor" condition embedded teacher-authored solution keys and mistake cards *in the prompt* for a fixed practice set. MindCraft's claimed architecture (living mastery graph + ingredient/bridge DAG + deterministic checkers) is not what Bastani tested.

- **KILLED:** "Guardrails $\Rightarrow$ learning gain vs. no-AI." Guarded GPT Tutor only matched the no-AI control — it did not beat it. The real product bar the lab sets is `solo_transfer_pass  $\geq$  no-AI control`, not merely "less harmful than raw ChatGPT." "Less crutch than ChatGPT" is table stakes, not a win condition.

- **SURVIVES (strengthened):** Unguarded generative practice access can genuinely harm solo exam performance (High confidence) — and the harm is invisible to the student in the moment. GPT Base students in Bastani did **not** perceive worse learning despite the exam collapse; GPT Tutor students perceived *better* exam performance despite showing no real gain. **Self-report / thumbs-up cannot certify learning** — a finding the lab treats as one of its hardest constraints on product analytics generally, not just on AI-tutor design.

Commercial wedge tested: Parent-facing: "If homework looks perfect and the quiz collapses, you met a crutch."* Never market "ChatGPT but nicer" or claim Bastani validates MindCraft's architecture.

2.6 Sycophancy — a distinct, deeper risk than hallucination

From chapters/33_ai_tutors_trust_sycophancy.md (Red Teamed 2026-07-26):

FACT: Sharma, Tong, Korbak et al. (2023/24, ICLR) — five SOTA assistants (Claude 1.3/2, GPT-3.5/4, Llama-2-70B-chat) show measurable **sycophancy**: matching user beliefs over truth. Human preference data and reward models sometimes *prefer* confidently-written sycophantic answers over correct ones — meaning RLHF itself can reward truth-sacrificing agreement. This is a structural property of how these models are trained, not a prompting bug.

FACT (novices can't detect it): Bo, Kazemitabaar, Deng, Inzlicht & Anderson (2025/26, CHI) — within-subjects study (n=24) of ML novices debugging with high- vs low-sycophancy chatbots: high-sycophancy caused misconceptions to *persist* and **~0% relative improvement** vs. **~+49%** for low-sycophancy — and most users **could not detect** the difference and rated both equally helpful. Perception metrics fail as safety monitors here too.

FACT (pressure-contingent): Kasneci & Kasneci (2026) — models that resist a simple context-switch attack can still capitulate under **authority** pressure ("my notes say I'm right") or **face-saving** pressure ("please don't tell me I'm wrong"). This is the specific failure mode most relevant to an anxious student pushing back on a correction.

Product bar set by the chapter: Solver/coach default = guarded help, never full-answer-first. `solo_transfer_pass ≥ no-AI` is the success bar. CI should run an anti-sycophancy eval suite (student-wrong / student-right / student-anxious / authority-pressure / face-saving prompts) and fail the build if a model affirms wrong math. One-sentence doctrine from the chapter: *"AI may speak; the graph must decide; the student must still do the decisive cognitive work — and the product must be willing to disagree."*

2.7 Trust repair after an AI is caught being wrong

From `chapters/60_trust_after_ai_hallucination.md`:

FACT (a genuinely surprising negative result): Liu, Du, Ma, Zhang & Yan (2024, IEEE THMS) — in a test-track automated-driving failure study (N=257), a verbal apology + explanation + promise **did not restore damaged trust**. This is a transfer-wounded finding (driving safety ≠ wrong algebra hint) but the lab explicitly keeps the conservative reading: **do not assume apology copy restores trust scores**.

FACT: Zhang et al. (2025/26, CHI) — for hallucination specifically (as opposed to bias or plain factual error), users showed **no clear preference** among rote, explanatory, or empathic apology styles — this is genuinely unsettled, not a solved UX problem.

Product doctrine (SAFE-REPAIR): detect → classify (competence miss vs. integrity miss, e.g. sycophantic affirmation of a known-wrong answer) → own it briefly → show a concrete correction path → require a re-attempt → escalate to a human only on defined triggers. Explicitly kills: "our AI doesn't hallucinate" marketing, rote/empathy-only apologies as the repair product, and thumbs-up as proof repair worked.

3. Market / Pricing / Positioning Research

Primary source: chapters/59_parent_wtp_experiments.md (Part LIX, SAFE-WTP), with market-sizing facts also in Part XXVII.

3.1 What's actually known about the market (FACT, not product-specific)

- Kim, Goodman & West (2024, CESifo/IZA) — U.S. private tutoring **centers** more than tripled (~3,000 → ~10,000) from ~1997–2022; locations concentrate heavily in **high-income, high-parent-education** areas (>half in the top income quintile).

- Same source: ~6–7% of U.S. families with kids 6–17 paid for tutoring in the past year (2022 CES data), averaging ~\$437 in months with a purchase; top 10%/5% of monthly spend near \$1,200/\$2,000. The lab is explicit this is a descriptive snapshot, **not a MindCraft price ceiling**.

- Bray (2010) / Bray & Lykins (2012) — "shadow education" (fee-based supplementary tutoring) is a well-documented global phenomenon, useful for competitive framing but not a US pricing guide.

Commercial implication the lab draws: there is real paid demand adjacent to MindCraft, but it's **skewed toward income/education-privileged families** — pricing strategy that assumes one national number will both miss the premium segment and reproduce shadow-education inequality if not designed against it deliberately.

3.2 The methods correction — this is the most reusable part of this chapter

FACT: Stated "how much would you pay?" (Likert/open-ended) WTP research is methodologically weak. Murphy, Allen, Stevens & Weatherhead (2005) meta-analysis:

hypothetical-vs-actual WTP has a median ratio of ~**1.35**, severely right-skewed — people say they'd pay more than they will. Choice-based methods (discrete-choice / conjoint, forcing a trade-off against a real "neither" option and a price) recover relative preferences more reliably than open-ended questions, per the Louviere/Hensher/Swait and McFadden random-utility-theory lineage.

What this kills for MindCraft specifically: treating survey dollars as a revenue forecast, treating "parents only buy score spikes" or "parents buy identity" as settled without a real trade-off study, and shipping streak/minutes/unlimited-AI as the default parent value proposition without it having actually won a choice experiment.

3.3 The proposed attribute set (shippable, not aspirational)

The chapter's design rule: *every CBC attribute level must map to a screen or ops cost MindCraft could actually ship in 90 days.*

Attribute	Levels drafted
Diagnosis depth	vague "needs practice" / concept weakness list / bridge+format gap map
Evidence report	score/minutes only / FEI pack (retry, challenge-seeking, transfer) / FEI + honest plateau note
Human accountability	none / weekly async tutor note / 2× live 25-min / 4× live
AI help policy	unlimited chat / guarded pedagogy wrap / attempt-first then cards
Practice architecture	streak gamification / mastery climate soft-wrong / graded exposure ladder
Price	e.g. \$49 / \$99 / \$179 / \$299/mo (calibrate to segment)
Outside option	"Neither — keep free Khan/GPT/status quo"

HYPOTHESIS (medium confidence): diagnosis depth + FEI report + modest human accountability will beat unlimited AI and streak theater **in the anxious-middle segment specifically** — the score-chaser segment may overweight timed-test simulations instead. The lab is explicit this must be measured per-segment, not averaged into one "parent persona."

3.4 What this chapter explicitly kills

Likert-style WTP as pricing research; survey dollars as revenue; "parents only buy scores" as a settled claim; streak/minutes/unlimited-AI as a default value prop without it winning a real trade-off; any guaranteed-ACT-point language in a pricing

tier; averaging distinct parent segments into one persona.

3.5 District / B2C(school) procurement as its own go-to-market motion

From chapters/73_district_procurement_privacy_gtm.md (Part LXXIII, SAFE-PROCURE) — relevant if MindCraft ever sells to schools rather than only direct-to-parent:

FACT: U.S. K-12 district purchases are gated by data-privacy agreements (DPAs), not by a "FERPA compliant" badge or a teacher's click-wrap acceptance of terms of service — the U.S. Dept. of Education's own PTAC guidance explicitly warns that teacher click-wrap can bind a district to terms that conflict with district policy and law. The **SDPC National Data Privacy Agreement (NDPA)** is the actual industry-standard contracting rail (13,000+ districts reference it).

State overlays are not optional: California Ed Code §49073.1 mandates specific contract clauses (no targeted advertising on pupil data, deletion certification, etc.); Illinois SOPPA similarly regulates "operators." A national privacy page that ignores these fails the first serious CA/IL review.

Commercial implication: the lab frames the whole privacy/DPA packet as a GTM artifact — "district revenue is gated by a *trust packet*, not by pedagogy claims."

Kills: FERPA badge as proof of readiness, free-classroom-as-viral-GTM bypassing the same approval a paid tool would need, and any biometric/emotion-camera feature as a "differentiator" (it's a poison pill for procurement, independent of whether it's a good idea pedagogically — see §4 in the SAFE-PRIVACY doctrine).

4. North Star / Metrics — instrumentation depth beyond the known summary

The already-known summary (retry_120s / challenge_accept / transfer_pass / solo_transfer_pass / HID) is accurate as the headline. Part XXI (Metrics Dictionary) goes much further than that summary suggests — it's the single largest structured artifact in the Constitution, cataloguing **~100+ named "banned as North Star" metrics* (*nearly one per SAFE-* rule — e.g. "Focus Score™," "Struggle Score™," "Curiosity Score™") and a matching set of real leading-indicator event names the lab actually wants instrumented. This is far more exhaustive than worth reproducing here; if you're building analytics, read Part XXI directly (grep "^## XXI" -A 5 in the Constitution, or the full section around line 1723).

The handful of leading indicators worth knowing without going deeper:

Metric	Definition	Why it exists
retry_120s	New attempt on same/isomorphic item within 120s of a soft-wrong	Persistence under safety
challenge_accept (+ challenge_motive)	Chose a harder level when offered, tagged mastery/appearance/normative	Challenge-seeking, motive-gated so gaming doesn't count
transfer_pass / solo_transfer_pass	Correct on a varied item after a mastery mark / with AI or Solver closed	Anti-fake-mastery / anti-crutch
ai_reveal_rate	Frequency of full-answer/unguarded reveals	Crutch exposure
hint_binge	≥3 hints without an independent solve	Gaming / helplessness signal

Decision rule (HYPOTHESIS, Part XXI.4): Ship changes that raise retry_120s and mastery-motive challenge_accept **without** raising hint_binge or ai_reveal_rate, and without dropping transfer_pass/solo_transfer_pass. Prefer delayed mixed-set accuracy over blocked-session accuracy as the read on whether something actually worked.

5. Product Rules Reference (SAFE-* doctrine, representative sample)

159 SAFE-* rules exist in total. Below is a sample of ~35 of the broadest, most-likely-to-be-reusable ones — general product doctrine on identity, motivation, feedback, difficulty, format, and tutoring quality — pulled from the Constitution's own **Part XIV Red Team Dossier** (the "Kill #N" entries), which is itself already a condensed one-paragraph-per-rule summary. Deliberately **excluded** from this table: the WTP/SAFE-WTP and district/SAFE-PROCURE rules (covered in §3 above), and nearly all of the chapter-100–160 QA-operations cluster (see §7). Each source chapter is independently readable in chapters/ if you need the full evidence base.

SAFE code	Kills (the plausible bad idea)	Do instead	Source
SAFE-DD	"Harder is always better" / desirable difficulty that ignores anxiety	Equip → destake → dose → frame → measure; gate by segment	Ch 41
SAFE-COMPARE / HABIT / REWARD	Leaderboards, streaks, XP as identity-builders	Criterion feedback, cue-based practice, informational (not controlling) unlocks	Ch 42–44
SAFE-RESILIENCE	"Grit/Resilience Score™" / staying in the "danger zone"	Growth-zone framing + support recruitment; measure recovery & transfer, not a meter	Ch 45
SAFE-EXPECTANCY	Pygmalion labels / warm AI that just finishes the problem	Task briefs + concrete, informative, output-focused feedback (CIOF)	Ch 46

SAFE code	Kills (the plausible bad idea)	Do instead	Source
SAFE-CALIB	"Raise confidence" / Belief Score™ / confidence-alone as the lever	Item-level elicit → classify miss → tiered feedback; appropriate (not maximal) confidence	Ch 50
SAFE-ONTOLOGY	Graph file / RAG chat sold as a "diagnosis moat"	Diagnosis-before-dialogue; inspectable longitudinal state; LLM at the bookends only	Ch 67
SAFE-HITL	Warm human / hours-booked / "got it?" as tutor quality	Map-briefed tutors; prompt-more-than-pour; QA on FEI outcomes, not warmth	Ch 68
SAFE-PRIVACY	Emotion-AI / Anxiety Score™ / "FERPA compliant" treated as ethics	Self-authored, short-TTL affect check-in feeding pedagogy only; no biometric emotion recognition	Ch 69
SAFE-LABMETA	Page-count-as-science / "Bayesian update" without an actual check	Label every claim → red-team it → kill/wound/survive → demote copy accordingly	Ch 70
SAFE-TUTORGRAIN	Ivy/expert résumé treated as a proxy for tutoring quality	Trained near-peer default; structure beats pedigree; expert = escalation/coach-of-tutors	Ch 71
SAFE-GENQ	LLM-generated items shipped straight to the bank	Independent key verification before ship; hard drop-rate gate before scaling generation	Ch 72
SAFE-SCHED	"Expanding SRS" / "AI found your perfect interval" marketing	Horizon-matched spacing intervals; delayed first return; instrument before branding	Ch 74
SAFE-WORKFORCE	Tutor headcount / tenure-months as a quality proxy	Fidelity-over-tenure; continuous coaching; caseload-aware hiring gates	Ch 75
SAFE-FORGET	"Mastery fireworks" sold as durable competence	Age the evidence explicitly; delayed retrieval probes as the truth surface	Ch 76
SAFE-DURABLE	Cram/bootcamp packages marketed as durable identity	Dual-rail GTM: durable-learn track + a labeled, honest late "prove" track	Ch 78
SAFE-FBTIME	"Instant feedback AI" as an unqualified hero claim	Mode-conditional timing (diagnostic = hide; learn = micro-delay + self-explain; prove = labeled KR)	Ch 79
SAFE-BRIDGE	Bridge/edge <i>count</i> sold as "we connect everything"	Connection-first narrative only where the join is genuinely weak; remediate the relation itself	Ch 80
SAFE-COLD	Day-one "mastery fireworks" from an empty/seed-only state	Humble prior → visibly-labeled seed data → hide-correctness probes before any green	Ch 81
SAFE-FORMAT	Format/vessel <i>count</i> or "we teach every format" as a hero claim	Format as first-class evidence; conversion between formats is itself the learning object	Ch 82
SAFE-PDASH	Tonight-%, streaks, or shame-ranked parent portals as "trust"	Proof-age + a return agenda; autonomy-supportive framing, not control	Ch 83
SAFE-TALK	Talk-ratio-% or silence-as-Socratic as tutoring quality	Construction beats airtime — measure student constructive share, not raw talk-%	Ch 84

SAFE code	Kills (the plausible bad idea)	Do instead	Source
SAFE-STORYLOAD	"Immersion" / lore-depth treated as automatically pedagogical	Cognitive-load-budgeted narrative; cut lore for novices; story ≠ transfer by default	Ch 85
SAFE-PROOF	User-count / star-wall / vague testimonials as "learning proof"	Enactive, dated, named solo-competence artifacts a parent/student can actually inspect	Ch 86
SAFE-COVER	Item-count / "complete ACT bank" as a coverage claim	A concept×level coverage matrix with gaps named as gaps, not hidden	Ch 87
SAFE-FADE	Always-full-worked / never-fading Solver as "the" learning UX	Fade guidance by stage, not by default; attempt required before reveal	Ch 88
SAFE-HELP	Unlimited hints / "help NPS" as a learning proxy	Instrumental (not executive/answer-getting) help; contingent, staged hint access	Ch 89
SAFE-INSTRUMENT	DAU/streak/XP or a composite "FEI Score™" as THE North Star	Ship the actual FEI event set with co-primary gates (see §4); resist Goodharting any single metric	Ch 90
SAFE-HINT	Unlimited free "hard" (bottom-out) peeks as kindness	Soft hints before hard; a hard peek costs something (a prior attempt/construction)	Ch 91
SAFE-EXPLAIN	Longer AI explanations / "Explanation Score™" as quality	Short, principle-first; self-explanation required before the AI's full wrap	Ch 92
SAFE-RETRIEVE	"All struggle is productive" / never-stuck answer delivery	Classify the stall type (tip-of-tongue vs. blank freeze) before deciding how to help	Ch 93
SAFE-PF	Treating every wrong attempt as "productive failure"	Deliberately designed generate → consolidate sequencing, gated by prior knowledge	Ch 94
SAFE-APPRENTICE	"Guild" cosplay / modeling-as-lecture / permanent never-fading coaching	Model → coach → scaffold → fade, with required student articulation at each step	Ch 95
SAFE-ERRCLIMATE	"Psychological Safety Score™" / poster-only safe-space messaging	Stay, diagnose, scaffold, require re-attempt — destake the moment, never the standard	Ch 103
SAFE-HWHELP	"Homework Completion Score™" / Chegg-style full-solve-and-done	Dual-rail: Help (instrumental, deadline-driven) vs. Practice (the actual FEI gym); assisted ≠ solo-ready	Ch 104
SAFE-ROI	"AI replaces tutors" / hours-booked as the tutoring KPI	Treat tutor minutes as genuinely scarce inventory; spend them on joins/repair/witness, not reps AI can do	Ch 99

Where the rest live: SAFE-WTP (pricing methodology) and SAFE-PROCURE (district GTM) are covered in §3. The remaining ~120 rules span narrower UI-copy and instrumentation specifics (chapters 96–99 and much of 100+) plus the QA-operations cluster in §7. The Constitution's Part XX (grep "Do not out-" MINDCRAFT_RESEARCH_

CONSTITUTION_v1.md`) has a single dense paragraph listing literally every SAFE-* code by name if you need the full index.

6. Founder Questions / Open Problems / Roadmap (condensed, Parts XI–XIII)

Founder Questions (the lab explicitly refuses to answer these for you)

1. If ChatGPT-style tutors become "good enough," what *relationship* do parents still buy?
2. Is MindCraft building for the student's identity — or for parents' anxiety about scores? Is that a real tradeoff or a false one?
3. Would the team accept lower short-term accuracy for higher long-term challenge-seeking?
4. Is the story-world layer load-bearing, or is gap-diagnosis + human tutor enough on its own?
5. What would falsify the identity-transformation thesis within 90 days?

Long-term Roadmap (research horizons, not a build plan)

- **H1 (now):** Prove the FEI loop actually moves retry behavior and challenge-seeking.
- **H2:** Prove identity language + the knowledge map produce durable self-concept change.
- **H3:** Prove human tutor + AI diagnosis beats AI-alone on persistence and transfer.
- **H4:** Only generalize beyond math (e.g. science identity) once the math-identity mechanism is actually solid — not before.

Open Problems (unresolved, named explicitly as unresolved)

1. The causal path from "narrative history of math" to identity change is under-identified — the lab doesn't know if it's real.
2. How to build relatedness/belonging at scale without it becoming creepy social surveillance.
3. How to measure identity change without demand effects (students telling you what they think you want to hear).
4. How to prevent gamification mechanics from colonizing meaning once they exist in the product at all.

5. AI's fluent-but-wrong failure mode vs. building real trust calibration in teenagers specifically.

6. Equity: does the story-world approach privilege culturally specific narratives in a way that disadvantages some students?

7. The QA-Panel-Ops Cluster (chapters ~100–160) — exists, mostly not your problem

Roughly the back third of the chapter list (SAFE-REVOKE through SAFE-AMBSEAL / SAFE-MIXREFRESH, chapters ~148–160 most narrowly, with related operational chapters starting as early as ~120) is **not** general product/learning-science research. It's a detailed internal operations doctrine for running a ****human grading/rating QA panel**** — dual-rater agreement thresholds, gold-standard test-item rotation and tip-resistance, rater-pool surge-hiring and demobilization, and SLAs for resolving "poison" (bad) labels. It reads as if the lab modeled this on quality-assurance practices from clinical/medical rating panels and standardized testing (mammography double-reading, CAT-exam item security, etc.) and applied the same discipline to MindCraft's own content-verification pipeline.

It's real, internally consistent doctrine — but genuinely narrow. Only worth reading if you are actually building or operating a human content-rating/verification pipeline (e.g., scaling the generated-question verify step, or standing up a dual-rater QA process for AI-generated figures/items). Start at `chapters/144_second_rater_thresholds_figured_c4.md` if that's your task; otherwise this cluster can be safely skipped.

8. What this document deliberately leaves out

This summary read the competitive/market chapters, the AI-trust chapters, the full Red Team Dossier (Part XIV, all 114 Kill entries), the Metrics Dictionary (Part XXI), and the Founder Questions/Roadmap/Open Problems sections directly from the Constitution — not a re-derivation. It did **not** re-read the ~70 general learning-science chapters (spacing, interleaving, self-explanation, cognitive load, etc., chapters 24–99 individually) beyond what's captured in the Kill-list summaries above; those are real and citation-backed but are the deep pedagogy layer beneath the

product rules, not typically what a new agent needs to start shipping. If you're about to build a *specific* feature (e.g., the hint economy, worked-example fading, spaced review scheduling), go read that feature's specific chapter directly — the table in §5 tells you which one.